



Kathan Rajendrasinh Zula

Roll No.: B23CM1063

B.Tech in Artificial Intelligence and Data Science

Indian Institute of Technology, Jodhpur

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentile	Year of Completion
B.Tech. in AI & Data Science	Indian Institute of Technology Jodhpur	8.19/10.0	2023-2027
Senior Secondary (XII)	GHSEB Board	90.94%	2022
Secondary (X)	GSEB Board	99.66%	2020

PROJECTS

- **High-Performance RAG API for Document Intelligence** Jul-2025 - Aug-2025
Tech Stack: Python, FastAPI, LangChain, OpenAI GPT-3.5, Pinecone, Docker
 - Engineered a scalable **RAG pipeline** with FastAPI and RESTful API endpoints, ensuring reliable enterprise document intelligence.
 - Implemented **dual-layer caching in Pinecone**, reducing query latency by 40% and improving system throughput.
 - Achieved **80% factual accuracy** in document-based Q&A by optimizing semantic chunking and structured prompt design.
- **Bias-Resilient Image Classification Pipeline** Jan. 2025 – Apr. 2025
Tech Stack: Python, PyTorch, ResNet-18, Knowledge Distillation, Computer Vision
 - Engineered an image classification model using **ResNet-18** to accurately classify highly biased datasets exhibiting severe age-gender correlations.
 - Mitigated age-feature reliance via heavy data augmentation (random cropping, color greying), **minority sampling**, and structural regularization (**Dropout, DropConnect**).
 - Executed **incremental knowledge distillation** utilizing a teacher model and **KL divergence loss**, achieving an impressive **82% accuracy** on the heavily biased dataset.
- **GPS Navigation System with Real-Time Pathfinding** Sep. 2025 – Dec. 2025
Tech Stack: Python, Flask, OpenStreetMap API, JavaScript, HTML/CSS
 - Designed and implemented a web-based GPS navigation system with **dynamic pathfinding** using Dijkstra's Algorithm.
 - Integrated **OpenStreetMap API** for geospatial data, live map visualization, and real-time traffic updates.
 - Built a responsive frontend with geolocation support, distance calculation, and interactive navigation features.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, JavaScript, SQL, Bash
- **AI/ML & RAG:** Generative AI, LLMs (GPT-3.5/4), LangChain, Pinecone, OpenAI API, Semantic Chunking, NumPy, Pandas, Scikit-learn
- **Web & API Development:** FastAPI, Flask, Asynchronous Programming, RESTful Microservices, Node.js, React.js
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite3
- **Tools & DevOps:** Git, Docker, Postman, Railway, Google Colab, VS Code

ACHIEVEMENTS

- **Hackathon** Advanced to **4th elimination round at Bajaj Finserv HackerX 6.0** by building an enterprise-grade RAG Document Intelligence API.
- **Competitive Programming** Peak **LeetCode rating 1563**; ranked **2142** in Weekly Contest 445.
- **JEE Mains (2022)** Secured **99.09 percentile** among 900,000+ candidates.
- **KVPY (2021)** Qualified with an All India Rank of **4890** in SX stream.

RELEVANT COURSEWORK

- Data Structures & Algorithms, Design and Analysis of Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Understanding, Speech Understanding, Data Engineering, Optimization in Machine Learning, Probability & Statistics, Linear Algebra, Maths for Big Data, Principles of Computer Systems 1 & 2, Human-Machine Interaction, Data Visualization

POSITIONS OF RESPONSIBILITY

- **Assistant Head Akshar** - The Literary Arts Club, IIT Jodhpur Aug. 2024 – Dec. 2024
 - Coordinated event planning and managed a team of volunteers, ensuring smooth execution of literary activities and workshops.